

The Blind Spot Paradox

The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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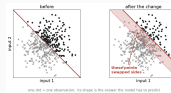
The Blind Spot Paradox: when a self-repairing model destroys the evidence its own monitor needs.

The Blind Spot Paradox

└ Concept drift: the rule changes, the data does not

First, the phenomenon. A model learns a rule: this input goes with that answer. Then the world changes and the rule no longer holds. That is concept drift, and here is the cleanest version of it, the one we simulate. Two inputs, a straight boundary, a label on each side. The change moves the boundary, and nothing else. Only the shaded band has swapped sides, and a model trained before is now wrong on it.

Concept drift: the rule changes, the data does not

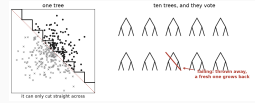


A model trained on the left keeps predicting, confidently, and is now **wrong** on the shaded band.

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└ Defence one: a model that repairs itself

The first defence repairs itself. Its building block is a decision tree: it cuts the picture straight across, so a slanted boundary can only be approximated by a staircase. One tree is fragile, so ten of them vote. That is the Adaptive Random Forest, the model we study. A tree whose answers go wrong is thrown away, and a fresh one grows in its place. Ten trees, replaced one at a time.



This is the **Adaptive Random Forest**. Ten trees, replaced one at a time.

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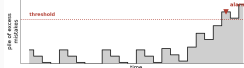
└ Defence two: a watchdog that raises the alarm

The second defence comes from a different community. A watchdog keeps a pile of the model's excess mistakes: it falls back to zero while the model does well, and grows until it crosses a threshold and calls a human. That alarm is the only thing a human ever sees. Two communities, two tools, never measured against each other.

Defence two: a watchdog that raises the alarm

the model assesses, one observation at a time. (x = wrong)

x o o o x o o o x o o x o o x o o x o o x o o x



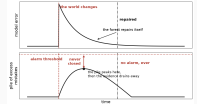
Two communities, two tools, *never measured against each other.*

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└ Put them together, and they cancel out

Put the two together, and they cancel out. The boundary moves, the mistakes begin, the pile starts climbing. But the forest repairs itself, the mistakes stop, and the pile drains back to zero without ever reaching the threshold. It is a race between a repair and an alarm, and the repair wins it every time. The better the model fixes itself, the more surely the alarm stays silent.

Put them together, and they cancel out



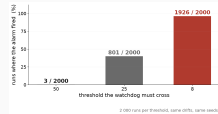
The repair has a date. The alarm never gets one. (schematic)

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└ Why not just lower the threshold?

Does the alarm ever fire? That depends on the threshold. We re-ran the benchmark at three settings, two thousand runs each. At the highest it fired three times out of two thousand. Lower it, and it fires almost every time. So why not lower it? Because a low threshold also fires when nothing happened, and an alarm that cries wolf gets switched off. The setting quiet enough to be trusted is the one that never sees the drift.

Why not just lower the threshold?



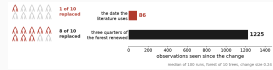
A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

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└ Our contribution: when is the forest actually repaired?

To claim the repair wins that race, you need a date for the repair, and everything the paper concludes rests on that single date. Here is the one it uses: the moment the very first tree out of ten is replaced. Wait until three quarters of the forest has actually been renewed, and you wait roughly ten times longer. So that stopwatch times the start of a repair, not its end, and it is wrong always in the same direction.

Our contribution: when is the forest actually repaired?



The accepted date fires at the first tree. It times the start of a repair, not its end.

The Blind Spot Paradox

└─Where we stand

Where we stand. The paradox is real and reproducible. The clock the field relies on is biased, always the same way, so every conclusion drawn from it inherits that bias. What remains is to turn this into advice a practitioner can use. Thank you.

- The paradox is **real and reproducible**, and we can say when the evidence falls short of what the alarm needs.
- The clock the field relies on is **biased, always the same way**. Every conclusion drawn from it inherits that bias.
- **Next:** watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.